

CMPE2150 SA 05

Relay and SSR Circuits

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24 Sep 2024

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Omron G2R-2-DC12

The first set of questions in this assessment are based upon your knowledge of transistor-controlled relay circuits and information provided in the manufacturer's datasheet for the relay in your kit, the [Omron G2R-2-DC12](#).

Note that this datasheet covers a number of devices, so you need to make sure you're accessing the correct part number when accessing the specifications.

1. Dielectric strength refers to _____.
 - a. the strength of a dielectric material, expressed as a function of aging due to repetitive arcing between terminal contacts.
 - b. the maximum voltage that may be applied across the terminals of a relay while they are open.
 - c. the maximum voltage that may be applied between the coil and the contacts before a resulting electric current will flow between them.
 - d. a measure of the capacitance of the polarizing electrostatic field that develops between switch contacts while they are open.
2. What is the basic dielectric rating of the G2R-2-DC12 relay, in volts? _____
3. The G2R-2-DC12 relay in your kit has a fully-sealed enclosure. _____ (True or False)

4. What type of protection does the enclosure of the G2R-2-DC12 relay provide? _____
 - a. prevents ingress of solder flux during soldering
 - b. prevents damage due to corrosive atmospheric gases
 - c. prevents protection from electromagnetic field flux
 - d. prevents influx of cleaning solvent during immersion cleaning
5. How many circuits can the G2R-2-DC12 control simultaneously?

6. For the G2R-2-DC12 relay, how many circuit elements can each relay switch be connected to? _____
7. Which statement describes the operation of the contacts in the G2R-2-DC12 relay?
 - a. all contacts are normally open
 - b. all contacts are normally closed
 - c. each switch has a normally-open contact and a normally-closed contact
8. What is the nominal coil voltage for the G2R-2-DC12 relay?

9. For the G2R-2-DC12 relay, what type of power should be supplied to the relay coil? _____ (AC or DC)
10. For the G2R-2-DC12 relay, once the relay is activated, what can the voltage be reduced to without the relay releasing, in volts?

11. What is the coil current rating of the G2R-2-DC12 relay, in milliamps? _____
12. What is the rated coil resistance of the G2R-2-DC12 relay?

13. The specified coil voltage and current accurately reflect the expected coil resistance for the G2R-2-DC12 relay. _____ (True or False)
14. If a 2N3904 is used to activate the 2GR-2-DC12 relay, in microamps, the Base current must be greater than _____ μA .
15. If a ZVN2110A FET transistor is used to activate the coil of a G2R-2-DC12 relay _____:
 - a. a diode should be placed, reverse biased, across the coil
 - b. a series resistor should be installed at the Gate
 - c. a pull-down resistor should be installed at the circuit input

- d. all of the above
16. If a ZVN2110A FET is used to activate a G2R-2-DC12 relay, the series Gate resistor should be designed to handle the coil current. _____ (True or False)
 17. What is the rated current-carrying capacity for the contacts of the G2R-2-DC12, in amps, with a resistive load? _____
 18. From the performance graphs for the G2R-2-DC12, what is the current handling capacity of the contacts, in milliamps, for a resistive load, if the supply voltage is $100V_{DC}$? _____
 19. If a G2R-2-DC12 relay is used to switch a 120 V circuit, the transistor used should be rated to over 120 V. _____ (True or False)
 20. The G2R-2-DC12 can be used as a switch for both DC and AC circuits. _____ (True or False)